Modulus and Argument

Key Points

The modulus and argument of a complex number, $z = x + yi$ is

$$|z| = \sqrt{x^2 + y^2}$$

$$Arg(z) = \tan^{-1}\left(\frac{y}{x}\right)$$

When $x < 0$ you need to alter the value on the calculator

- $x < 0, y > 0$ add π
- $x < 0, y < 0$ subtract π

The argument goes between

$$-\pi < Arg(z) \leq \pi$$

Basic Skills

Q1. Calculate the modulus of the following complex numbers

a) $3 + 4i$

b) $3 - 4i$

c) $5i - 1$

Q2. Calculate the argument of the following complex numbers

a) $3 + 4i$

b) $1 - 9i$

c) $10i + 1$

Q3. Explain where the formula's for the modulus and argument come from

Competency Skills

Q4. a) Plot the complex number $z = -3 + 4i$ on an argand diagram

b) Calculate the modulus and argument of the complex number

c) Using your argand diagram show what the modulus and argument represent for z

Q5. Calculate the modulus and argument of the following complex numbers

a) $9 - 7i$ b) $-4 - 3i$ c) $3i - 4$ d) $\frac{4}{3} - i$ e) $-\frac{1}{3}(i - 3)$

Q6. While we may measure the argument between $\pi < Arg(z) \leq \pi$, someone tells you they got the argument to be $\frac{3\pi}{2}$ can you change their value to fit with our choice of measuring arguments

Advanced Skills

Q7. Calculate the distance between the complex numbers $z = 4 - i$ and $w = -3 + i$ by calculating $|z - w|$

Q8. Someone says that the modulus of the complex conjugate is equal to the modulus of the complex number, state if they are right or wrong, providing your reasoning

Q9. Using $z = x + yi$ to prove that $zz^* = |z|^2$

Q10. Can you describe the set of complex numbers such that they have $|z| = 1$ by drawing an argand diagram

Q11. Can you describe the set of complex numbers such that they have $|z - (2 + i)| = 1$ by drawing an argand diagram

Extension

Q12. Describe the set of points on the argand plane such that $Arg(z) = \frac{\pi}{4}$

Hint: Think about what the argument means